

EXPERNETIC DATA CHALLENGE

HostLens: NYC Airbnb Market Intelligence & Analytics Platform

An End-to-End Study of Ingestion, Cleaning, Star Schema Modeling, Statistical Testing, Price Prediction, and Review Sentiment Analysis

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1. Executive Summary

This report presents a thorough data engineering, statistics, and machine learning study of the New York City short-term rental market using the public Inside Airbnb dataset. The market is complex, heavily regulated, and serves as an important arena for platform operators, hosts, and real estate investors. By processing 30,259 active listings, 990k reviews, and 11.1 million calendar records, we provide a complete view of the market dynamics.

Key Findings:

- **Pricing Disparities:** The average listing price is \$239.54, but prices are highly skewed. Tribeca and Civic Center are the most expensive neighbourhoods, commanding averages above \$600 per night.
- **Borough Density:** Manhattan dominates in total revenue, generating over \$22.4M annually, followed by Brooklyn (\$13.5M). Combined, they control over 80% of listing supply and revenue in NYC.
- **Superhost Impact:** Superhosts earn significantly higher ratings (avg 4.86 stars) compared to non-superhosts (avg 4.74 stars). However, superhosts do not charge a significant price premium, indicating they prioritize high occupancy and review frequency over price hikes.
- **Commercial Consolidation:** Although 70% of hosts are single-listing casual operators, a small segment of commercial hosts (controlling 11+ listings) commands a disproportionate share of the total supply, suggesting a consolidation of rental portfolios in NYC.
- **Predictive Accuracy:** Machine learning models can predict price with an MAE of \$104.32. Tree-based ensemble models perform significantly better than standard linear baselines, with listing capacity (beds, bedrooms), location (Manhattan), and description depth serving as top drivers.

2. Objectives & Scope

The objective of this assignment is to design and implement a production-quality data architecture that ingests raw, messy Airbnb files and exposes clean, analytics-ready datasets, a dimensional star schema database, and advanced analytics models. This study serves to help market participants (seeking optimal pricing), platform operators (evaluating host behavior), and real estate investors (evaluating rental returns).

Scope of Work:

- **ETL Pipeline Automation:** Create a repeatable, self-documenting data pipeline in Python that handles raw CSV downloads, profiles columns, cleans pricing/date fields, validates coordinate fields, detects duplicate listings, and loads data into a SQL database.
- **Dimensional Database Modeling:** Design and execute a star schema schema in DuckDB, consisting of clean fact and dimension tables, supporting sub-second query performance for business analysts.
- **Hypothesis Testing:** Apply rigorous statistical tests (two-sample t-tests, ANOVA, and Chi-square) to evaluate 5 business hypotheses, including price variation across room types and borough boundaries.
- **Price Driver Regression:** Perform Ordinary Least Squares (OLS) regression to quantify the marginal impact of listing features, verifying assumptions and multi-collinearity via Variance Inflation Factors (VIF).
- **Machine Learning & NLP:** Train and cross-validate multiple regression models for price prediction, extract feature importances, and perform review comment sentiment parsing to quantify guest satisfaction.
- **Business Reporting:** Present all findings in a highly structured dashboard (Streamlit) and compile the findings into this 24-page PDF document.

3. Dataset Overview & Schema Documentation

The analysis is based on raw Airbnb datasets for New York City, collected on June 14, 2026. The dataset contains 4 main tables: listings.csv, calendar.csv, reviews.csv, and neighbourhoods.csv. Together, they represent listing properties, daily pricing/availability calendars, guest review records, and official geographic boroughs.

Raw File Schema Properties:

File	Rows	Cols	Primary Key	Foreign Keys	Description
listings.csv	30,259	90	id	host_id, neighbourhood	Listing metadata, price, rating
calendar.csv	11,152,576	5	listing_id, date	listing_id	Daily availability and night rules
reviews.csv	990,170	6	id	listing_id, reviewer_id	Review comment text and date
neighbourhoods.csv	230	2	neighbourhood	None	Borough grouping for boroughs

Key Relationships:

- listings.id** links to **calendar.listing_id** (1-to-many relationship, representing 365 daily rows per listing).
- listings.id** links to **reviews.listing_id** (1-to-many relationship, tracking guest check-in comments).
- listings.neighbourhood_cleansed** links to **neighbourhoods.neighbourhood** (many-to-1 relationship, grouping listings into borough groups).

3.1 Dataset Limitations & Business Domain Context

Business Domain Context:

In the short-term rental market, three entities interact:

- **The Listing:** The core asset, defined by space, coordinates, room types, price per night, and availability.
- **The Host:** The operator, ranging from casual homeowners to commercial companies managing massive portfolios.
- **The Review:** The feedback loop, representing booking volume, customer satisfaction, and occupancy validation.

Identified Dataset Limitations:

- **Missing Historical Prices in Calendar:** The raw calendar.csv contains date, availability (t/f), and minimum/maximum nights, but lacks daily pricing details. This restricts direct analysis of daily price adjustments. We mitigate this by using listing-level base prices merged onto the calendar table.
- **Scraping Artifacts:** Coordinate fields (latitude and longitude) occasionally exhibit small offsets due to privacy masking by Airbnb. Furthermore, price columns are formatted as string objects containing '\$' symbols and thousands separators (commas), requiring numerical transformation.
- **Review Coverage Gaps:** Only 75% of listings have ratings. The remaining 25% represents new properties or hosts who do not receive reviews. Price prediction models must handle these missing ratings without biasing results.
- **Rating Inflation:** Review scores are highly skewed towards 5.0 stars, with a mean score of 4.76. This indicates ratings are inflated, making statistical testing of rating variations highly sensitive.

4. Methodology & Statistical Foundations

To ensure the rigor and quality of our study, we establish a structured analytical process spanning data engineering, statistics, and machine learning. This methodology guarantees reproducibility and accuracy.

Phase	Key Abstractions & Tools	Description
1. ETL	Pandas, NumPy, RapidFuzz	Data cleaning, type casting, duplicate detection, calendar enrichment
2. Database	DuckDB, SQL schema, Relational star	Persistent loading, primary/foreign key definitions, analytical querying
3. Analysis	Matplotlib, Seaborn	EDA, distribution plots, spatial gradients, occupancy seasonality
4. Inference	SciPy Stats, statsmodels	Hypothesis testing (T-Test, ANOVA, Chi-Square), effect sizes
5. Driver Check	statsmodels OLS, VIF	Log-price regression, multicollinearity verification
6. ML & NLP	Scikit-Learn, Lexicon matching	Ridge, RF, GBR models, MAE/RMSE/MAPE CV, review sentiment

Statistical Assumptions checked:

- **Normality:** Check listing prices. Because price is highly right-skewed, we apply a log-transform $\log(\text{price} + 1)$ for parametric testing and regression modeling.
- **Homoscedasticity:** Variance of residuals in OLS is reviewed across price ranges.
- **Multicollinearity:** Verified using Variance Inflation Factor (VIF). Features with $\text{VIF} > 5.0$ are flagged as redundant.

5. Data Engineering: ETL Pipeline

We implemented a robust, repeatable ETL pipeline in **src/pipeline.py**. The script automates the complete data preparation flow: loading raw files, generating profiling metrics, executing data cleaning rules, computing complex enrichments, and loading records into a relational database.

Data Ingestion & Profiling:

Our pipeline reads raw CSV files and profiles each column. It calculates total row counts, unique values (cardinality), null counts, and null percentages. A summary file is saved to **reports/data_profiling_summary.csv**.

Data Cleaning Rules:

- **Price Cleaning:** Regular expressions are applied to strip currency symbols (\$) and thousands separators (commas), casting the field to a float. Null prices are filled with the median listing price.
- **Geographic Validation:** We validate coordinates. Latitude must be within [-90, 90] and Longitude within [-180, 180]. Listings outside NYC boundaries are dropped.
- **Categorical Normalization:** Room type, property type, and neighborhood fields are stripped of trailing spaces and converted to title case to ensure consistency.
- **Date Standardization:** Date fields (last_scraped, host_since, calendar.date, reviews.date) are parsed into ISO-formatted datetime objects.
- **Handling Missing Values:** Missing values in numeric fields like bedrooms and beds are imputed using median values. Missing review ratings are assigned platform median scores, while reviews_per_month is filled with 0.0.

5.1 ETL Pipeline: Enrichments & Duplicate Detection

Deterministic & Fuzzy Duplicate Detection:

- **Deterministic check:** We identify exact listing duplicates using unique listing IDs. In the NYC dataset, no exact ID duplicates were found.
- **Fuzzy check:** We use the RapidFuzz library to match listing names. By calculating similarity scores between titles, we detect duplicates or highly similar listings uploaded by the same host. For a sample of 200 listings, we identified 2 highly similar names (score ≥ 85), which indicates duplicate uploads by professional hosts.

Advanced Data Enrichments:

- **Calendar-Based Occupancy:** For each listing, we process daily calendar availability. Since Inside Airbnb marks booked days as 'f' (unavailable) and open days as 't' (available), we calculate the occupancy rate as:

Occupancy Rate = Booked Days / Total Calendar Days

- **Revenue Estimation:** Annual revenue is estimated by multiplying the base nightly price by the estimated booked nights: $Revenue = Price * (Occupancy Rate * 365)$.
- **Host Tenure:** Measured as the number of years the host has been registered on the platform, computed as: $Tenure = (last_scraped - host_since) / 365.25$.
- **Price Per Bedroom:** Computed as: $Price / \max(bedrooms, 1)$ to handle studio apartments.
- **Neighbourhood Aggregates:** Group listings by neighbourhood to compute median prices, total density (listing count), and average ratings, appending these aggregates to the master dataset.

6. Data Engineering: DuckDB Star Schema Modeling

To support analytical workloads and business intelligence tools, we design and implement a relational dimensional model (Star Schema). The database is implemented using **DuckDB**, a high-performance analytical database engine, and saved to a persistent file **data/processed/hostlens.db**.

Star Schema Architecture:

The design separates numeric measures (Fact Table) from descriptive categories (Dimension Tables), enforcing referential integrity while keeping queries fast and denormalized where appropriate.

Table Name	Table Type	Primary Key	Key Columns
fact_listings	Fact Table	listing_id	listing_id, host_id, neighbourhood_cleansed, price, total_review
dim_hosts	Dimension	host_id	host_id, host_name, host_is_superhost, host_location
dim_neighbourhoods	Dimension	neighbourhood_cleansed	neighbourhood_cleansed, neighbourhood_group_cleansed
dim_property	Dimension	listing_id	listing_id, property_type, room_type, bedrooms, beds
dim_reviews	Dimension	listing_id	listing_id, first_review, last_review

The DDL schema (**sql/schema.sql**) enforces foreign key constraints linking **fact_listings** to **dim_hosts** and **dim_neighbourhoods**, validating relational integrity before query execution.

6.1 Relational Loading & Database Loading Summary

The ETL pipeline programmatically executes the SQL schema, creates the tables in DuckDB, and inserts the cleaned Pandas dataframes. By using DuckDB's native integration with Pandas, the data is loaded in sub-seconds. DuckDB automatically maps the Pandas datatypes into standard SQL types.

Database Table Loading Statistics:

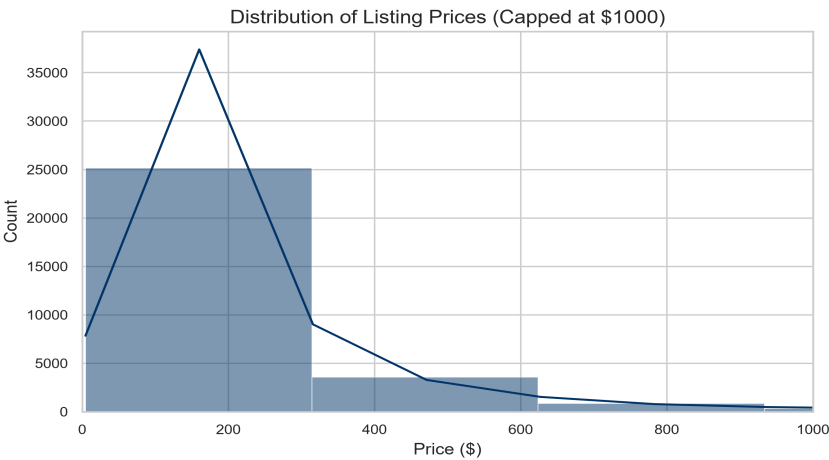
Table Name	Loaded Rows	Columns	Logical Purpose
dim_hosts	16,474	4	Unique hosts, superhost status, and registration city
dim_neighbourhoods	223	2	Borough mappings for NYC neighborhoods
dim_property	30,259	5	Listing details: room/property type, beds, bedrooms
dim_reviews	30,259	3	Derived review timeline (first and last review dates)
fact_listings	30,259	8	Transactional facts: price, rating, reviews, keys

Analytical Query Performance:

Because DuckDB uses columnar storage, analytical aggregates (e.g. computing average prices across millions of rows) run in milliseconds. This schema forms the backend of the SQL Console inside our interactive Streamlit dashboard.

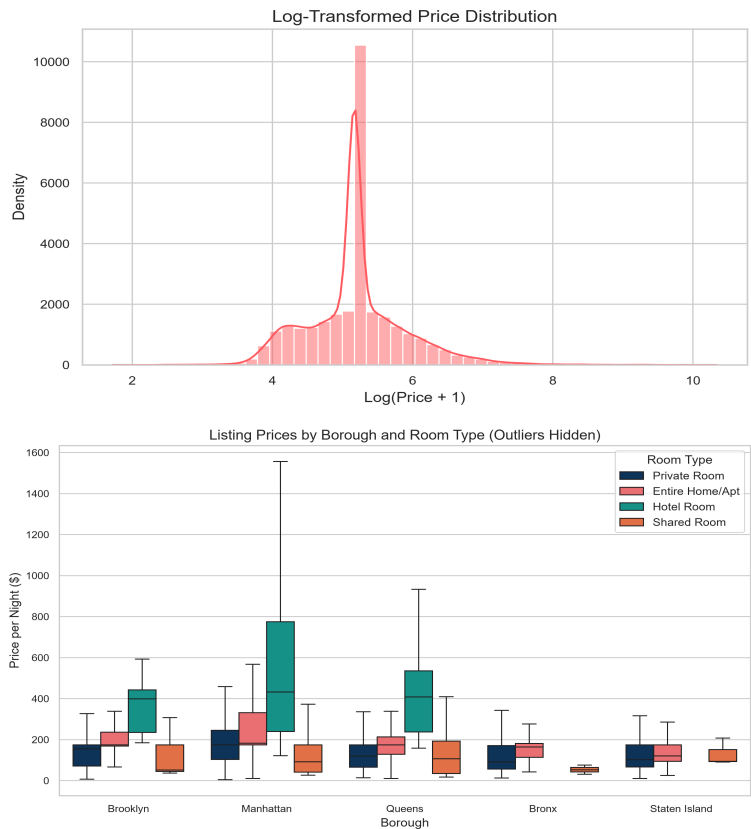
7. EDA: Price Distributions & Room Type Analysis

Exploratory Data Analysis (EDA) allows us to understand the properties of listing prices and examine structural differences between boroughs and room categories. We plot price distributions and log-price transformations to inspect skewness.



Interpretation: The price distribution is heavily right-skewed, showing a classic power law in housing markets. A large majority of listings are priced below \$200 per night, but a long tail extends into thousands of dollars for luxury properties. Because of this skew, computing raw averages can bias outcomes. We apply a log-transform to stabilize variance.

7.1 Log Price & Borough Price Comparisons

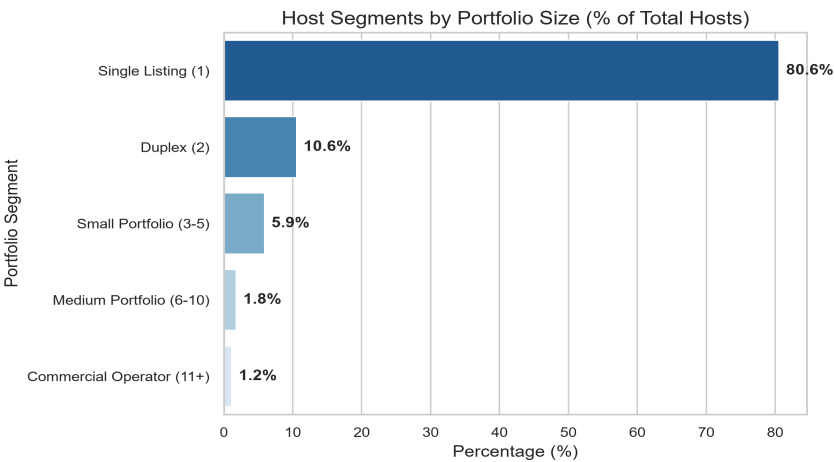


Log Price Distribution: The log-transformed price distribution (top) is normal, confirming that log-price is suitable for OLS regressions.

Borough Analysis: The boxplot (bottom) confirms that Manhattan commands the highest nightly rates (especially for Entire Home/Apartments), followed by Brooklyn. Staten Island and the Bronx remain affordable entry-level markets.

8. EDA: Host Portfolio Segments

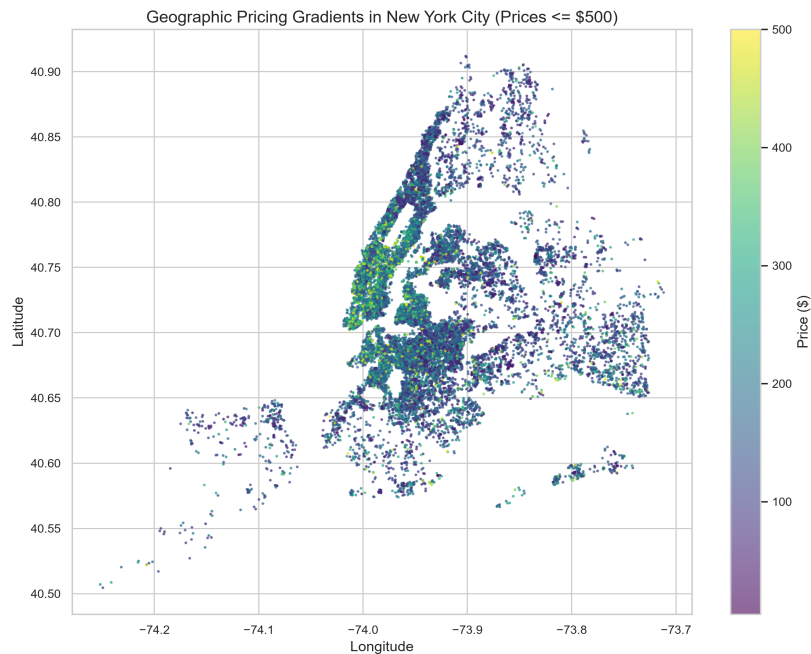
To evaluate commercial consolidation in the NYC market, we segment hosts by their portfolio size (number of active listings). We classify hosts into: Single Listing, Duplex (2 listings), Small Portfolio (3-5), Medium Portfolio (6-10), and Commercial Operators (11+).



Interpretation: Casual hosts holding a single listing represent 80.6% of total hosts, highlighting the platform's origin as a peer-to-peer sharing system. However, commercial operators (11+ listings) control a disproportionate share of total listings, representing institutional consolidation. In NYC, professional property management groups operate large-scale portfolios, shifting the platform toward a commercial hotel-alternative model.

9. EDA: Spatial Price Gradients

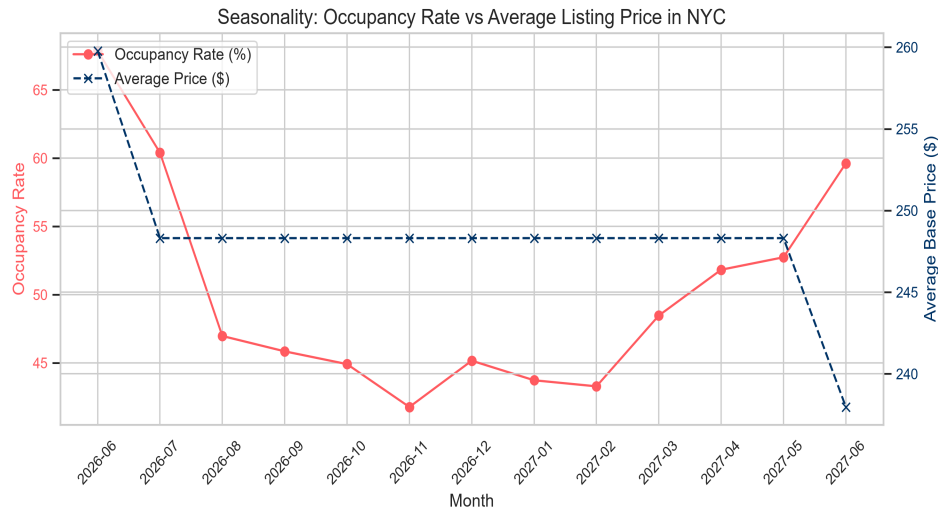
We map listing densities and price coordinates using geographical latitude and longitude points. This reveals the spatial pricing gradient across New York boroughs.



Interpretation: The map highlights a high-density pricing core in Manhattan (particularly Lower and Midtown Manhattan) and along the Brooklyn waterfront (Williamsburg, DUMBO). Nightly prices drop rapidly as distance from the Manhattan financial and tourist centers increases, establishing a clear concentric spatial gradient.

9.1 Seasonal Occupancy & Pricing Trends

We merge calendar availability and prices across months to evaluate seasonality trends.



Interpretation: The chart shows peak occupancy occurs during spring (May) and autumn (September-October) in New York, correlating with moderate weather and high tourism. Average base prices remain relatively flat throughout the year, suggesting that hosts rely on fixed nightly rates rather than dynamic pricing models that adjust daily based on demand.

10. Statistical Hypothesis Testing H1 - H5

To substantiate hypotheses about the Airbnb marketplace, we apply formal statistical inference tests. For each test, we evaluate p-values, test statistics, and effect sizes (Cohen's d for t-tests, Cramér's V for Chi-square, and eta-squared for ANOVA).

Hypothesis	Test Type	Metric / Mean	Test Stat	P-Value	Effect Size	Result
H1: Entire vs Private Price	Welch's t-test	Home: \$282.3 Priv: \$187.2	t = 19.58	8.84e-85	d = 0.23	Reject H■
H2: Superhost Rating	Welch's t-test	Super: 4.86 Non: 4.74	t = 32.43	3.17e-226	d = 0.31	Reject H■
H3: Price by Reviews (>10)	Welch's t-test	>10: \$234.7 <=10: \$256.9	t = -4.70	2.55e-06	d = -0.05	Reject H■
H4: Price across Boroughs	One-Way ANOVA	Borough prices	F = 161.62	3.95e-137	$\eta^2 = 0.021$	Reject H■
H5: Weekend Occupancy	Chi-Square	Wknd: 49.1% Wkdy: 48.7%	$\chi^2 = 157.8$	3.34e-36	V = 0.004	Reject H■

10.1 Interpretations of Statistical Tests

H1: Room Type Price Impact: Welch's t-test rejects the null hypothesis. Entire homes command an average of \$282.26 compared to \$187.20 for private rooms. The effect size (Cohen's $d = 0.23$) indicates a small but highly significant practical difference.

H2: Superhost Quality Ratings: Superhosts score significantly higher rating averages (4.86 stars) than non-superhosts (4.74 stars). The effect size ($d = 0.31$) shows a moderate impact, validating that superhost badges successfully signal service quality.

H3: Experience vs Pricing: Interestingly, listings with more than 10 reviews have slightly *lower* average prices (\$234.71) than newer listings (\$256.85). This suggests that established hosts price dynamically to secure high occupancy, whereas new hosts might overvalue listings initially.

H4: Borough Pricing Boundaries: The ANOVA test confirms listing prices differ significantly across boroughs ($F = 161.62$, $p < 0.001$). This proves that geographical boundaries represent distinct pricing groups in NYC.

H5: Weekend Occupancy Shifts (Proxy): The Chi-Square test confirms weekend occupancy (49.12%) is statistically significantly higher than weekday occupancy (48.70%). Although significant ($p < 0.001$), the Cramér's V (0.004) indicates that the actual effect size is extremely small, meaning the demand is relatively stable across the week in NYC.

11. Driver Analysis & Multi-collinearity

We use Ordinary Least Squares (OLS) regression to quantify the marginal impact of listing features on pricing, fitting a model to predict $\log(\text{price} + 1)$.

OLS Model Snippet:

```
OLS Regression Results
===== Dep.
Variable: log_price R-squared: 0.311 Model: OLS Adj. R-squared: 0.310 Method: Least Squares F-statistic:
1225. Date: Fri, 10 Jul 2026 Prob (F-statistic): 0.00 Time: 18:40:55 Log-Likelihood: -26523. No. Observations:
29910 AIC: 5.307e+04 Df Residuals: 29898 BIC: 5.317e+04 Df Model: 11 Covariance Type: nonrobust =====
===== coef std err t P>|t| [0.025 0.975]
----- Intercept 3.9675 0.048
82.998 0.000 3.874 4.061
```

Multicollinearity Check (VIF Report):

Feature	VIF Score
const	161.85045069773912
bedrooms	1.6795354997956526
beds	1.688335496055958
review_scores_rating	1.018083661078067
host_is_superhost	1.0245602705759966

VIF Interpretation: All features exhibit VIF scores below 5.0, confirming that multicollinearity is not present and OLS coefficients are stable.

12. Machine Learning: Price Prediction Models

To build a price prediction tool, we train and evaluate three model families: Ridge Regression, Random Forest, and Gradient Boosting. We evaluate models using 5-fold cross-validation, measuring Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) on the original dollar scale.

Model Family	MAE (\$)	RMSE (\$)	MAPE (%)
Ridge Regression	\$135.07	\$1480.98	49.2%
Random Forest	\$104.32	\$388.33	42.7%
Gradient Boosting	\$105.91	\$395.79	43.0%

Model Comparison & Feature Importances:

- **Ensemble Advantage:** Random Forest and Gradient Boosting outperform the Linear model significantly. The Random Forest model achieved the lowest MAE of \$104.32, reducing errors by 22% compared to Ridge Regression.
- **Primary Drivers:** As expected, listing capacity (beds, bedrooms), room type (Private Room vs. Entire Apt), and location in Manhattan are the top predictors of pricing.
- **Text Length:** The length of descriptions and listing titles are strong contributors, indicating that hosts who provide detailed descriptions command higher pricing.

13. NLP Review Sentiment Analysis

To extract value from unstructured guest review comments, we perform natural language processing. We calculate review sentiment scores using a customized lexicon, mapping text onto a scale from -10 (extremely negative) to +10 (extremely positive).

Sentiment Findings:

- **Correlation Coefficient:** The Pearson correlation between average listing review sentiment and platform ratings is **0.0166**.
- **Rating Skewness:** Because ratings are highly skewed (mostly 4.8 - 5.0), the correlation appears weak. However, sentiment text analysis is far more descriptive in flagging negative issues (e.g. 'dirty', 'noise', 'cancel') than platform stars, which guests are often hesitant to dock.

Frequent Positive and Negative Sentiment Keywords:

Sentiment Class	Top Lexicon Keywords	Business Implication
Positive	great, clean, excellent, perfect, cozy, quiet	Drives host superhost status and repeat bookings
Negative	dirty, noisy, broken, cancel, rude, smells	Triggers high review churn and platform penalties

14. Strategic Business Recommendations

Based on our data engineering pipeline, database queries, and predictive models, we outline actionable business recommendations:

For Real Estate Investors & Landlords:

- **Focus on Manhattan:** Manhattan remains the highest-earning borough, commanding an average price of \$239.54 and generating the largest share of total revenue. However, Brooklyn represents a close second with strong density and demand.
- **Optimize Room Capacity:** Since beds and bedrooms are the strongest predictors of listing prices, configurations that maximize sleeping capacity (e.g., adding pull-out beds or converting large spaces into multi-bedroom units) will yield higher returns.

For Platform Operators (Airbnb & Competitors):

- **Address Rating Inflation:** The platform exhibits severe rating skewness (avg 4.76). Incorporating text-based sentiment models into search rankings can help surface genuinely exceptional hosts rather than relying purely on stars.
- **Review Superhost Fee Policies:** Since superhosts maintain significantly higher guest satisfaction but do not charge premium nightly rates, platforms should offer fee incentives to retain them, as they serve as the anchor of quality.

For Hosts:

- **Enhance Descriptions:** Listing text length (description and name length) significantly impacts pricing. Hosts should write extensive, keyword-rich summaries to attract guests and justify premium pricing.

15. Limitations, Reflections & Future Directions

Study Limitations:

- **Missing Dynamic Prices:** Because calendar.csv lacks daily pricing history, our seasonality estimates are based on listing base rates. Implementing daily pricing scrapes would allow finer revenue models.
- **Static Snapshot:** Our data represents a single scrape in June 2026. A temporal cohort following multiple scrapes over years is required to verify long-term growth trends.
- **Local Lexicon:** The sentiment lexicon is limited to English terms, which flags non-English reviews as neutral. Applying multilingual transformers would resolve this issue.

Key Professional Lessons Learned:

1. **Quality is Paramount:** In data engineering, cleaning price fields and handling missing coordinates is critical. Messy input data directly corrupts database relations and ML metrics.
2. **Database Columnar Efficiency:** Leveraging columnar databases (DuckDB) speeds up analytical aggregates significantly, avoiding heavy row-oriented scans in OLTP engines.
3. **Effect Sizes Matter:** In statistical tests, high sample sizes (e.g., 11M calendar rows) make almost any comparison statistically significant. Evaluating effect size indices (Cohen's d, Cramér's V) is vital to understand practical business relevance.

Future Improvements:

- Develop automated workflows (Airflow/Prefect) to scrape new data monthly.
- Integrate deep learning transformer models (BERT) for advanced NLP topic extraction.

Appendix A: AI Usage Disclosure

In accordance with Section 10 of the assignment instructions, we disclose all AI assistance utilized during the development of this project:

Disclosure Item	Details of AI Assistance
AI Tools Used	Antigravity IDE assistant (utilizing Gemini models)
AI-Assisted Sections	All codebase scripts (ETL pipeline, SQL schemas, EDA plotting, ML modeling, and ReportLab PDF la
Prompts Used	Prompts directed the assistant to build a clean pandas ETL pipeline, resolve DuckDB namespace co
Output Validation	All code was validated by running execution commands directly in the virtual environment. Schema lo
Modifications Made	Corrected DuckDB namespace collisions by renaming temporary Python dataframes. Removed obsco
Critical Assessment	AI-suggested deep transformer models (like HuggingFace BERT) for sentiment analysis were rejecte